



Southeast University

Introduction to Software Engineering

Wang Lulu, Liao Li

lliao@seu.edu.cn

Outline

Part 0. Overview

Part 1. Software Processes

Part 2. Modeling

❖ **Chap 7. Understanding Requirements**

❖ **Chap 8-10. Requirements Modeling**

❖ **Chap 11. Design Concepts**

❖ **Chap 12-14. Design Methods**

Part 3. Quality Assurance

Part 4. Software Project Management

Chap 8. Scenario-Based Methods

❖ 1. Requirement Analysis

- Goals
- Rules of Thumb
- Domain Analysis
- Elements of Requirements Analysis

❖ 2. Scenario-Based Modeling

- Create an initial use case
- Refine the initial use case
- Review the use-case

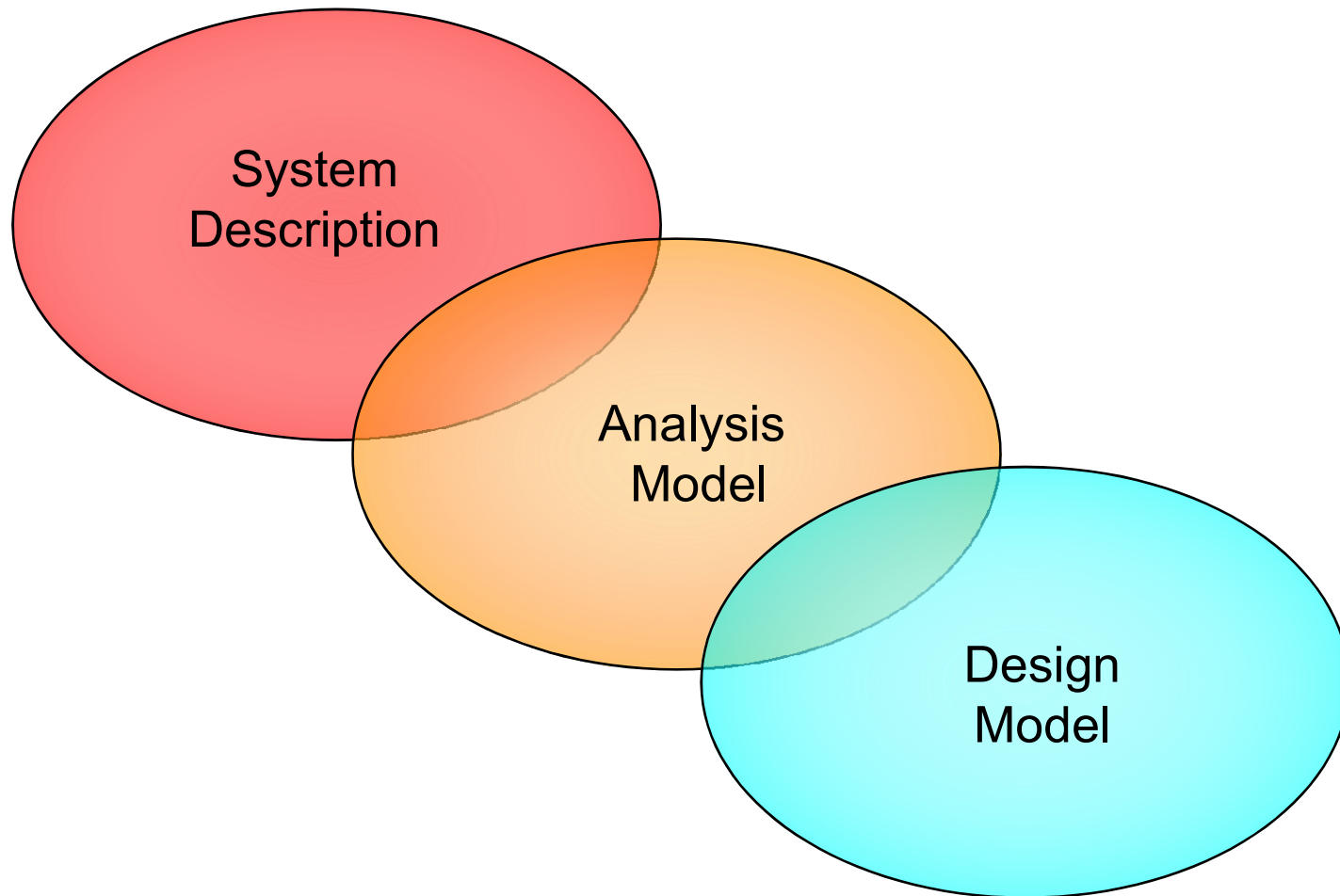
1. 需求分析

- 目标
- 经验法则
- 领域分析
- 需求分析的元素

2. 基于场景的建模

- 创建初步用例
- 提炼初步用例
- 审查用例

A Bridge



需求分析

Requirements Analysis

❖ Requirements analysis

- specifies software's **operational characteristics**.
- indicates software's **interfaces** with other system elements .
- establishes **constraints** that software must meet.

❖ Requirements analysis allows the software engineer (called an *analyst* or *modeler* in this role) to:

- elaborate on basic requirements established during earlier requirement engineering tasks.
- build models that depict user scenarios, functional activities, problem classes and their relationships, system and class behavior, and the flow of data as it is transformed.

需求分析

- 需求分析指定软件的操作特性。
 - 指出软件与其他系统元素的接口。
 - 建立软件必须满足的约束条件。
- 需求分析允许软件工程师（在此角色中称为分析师或建模师）**
- 详细说明在先前需求工程任务中建立的基本需求。
 - 构建描述用户场景、功能活动、问题类及其关系、系统和类行为以及数据流转模型。

Requirements Modeling

❖ Scenario-based

- system from the user's point of view

❖ Data

- shows how data are transformed inside the system

❖ Class-oriented

- defines objects, attributes, and relationships

❖ Flow-oriented

- shows how data are transformed inside the system

❖ Behavioral

- show the impact of events on the system states

- 基于场景

- 从用户的角度看系统

- 数据

- 显示数据在系统内部的转换方式

- 基于类

- 定义对象、属性和关系

- 基于流程

- 显示数据在系统内部的转换方式

- 行为

- 显示事件对系统状态的影响

Requirement Modeling

❖ 需求模型必须实现的三个目标:

- 1) 描述客户需要什么
- 2) 为软件设计奠定基础
- 3) 定义在软件完成后可以被确认的一组需求

Rules of Thumb （分析的经验原则）

- ❖ The model should focus on **requirements that are visible within the problem or business domain**. The level of abstraction should be relatively high.
- ❖ Each element of the analysis model should add to an overall understanding of software requirements and provide insight into the information domain, function and behavior of the system.
- ❖ Delay consideration of infrastructure and other non-functional models until design.
- ❖ **Minimize coupling** throughout the system.
- ❖ Be certain that the analysis model **provides value** to all stakeholders.
- ❖ Keep the model as **simple** as it can be.

- 模型应聚焦于问题或业务领域内可见的需求。抽象级别应相对较高。
- 分析模型的每个元素应增加对软件需求的整体理解，并提供对信息域、功能和系统行为的洞察。
- 延迟对基础设施和其他非功能模型的考虑，直到设计阶段。
- 在整个系统中最小化耦合。
- 确保分析模型为所有利益相关者提供价值。
- 尽可能保持模型的简单性。

Domain Analysis

Software domain analysis is the identification, analysis, and specification of common requirements from a specific application domain, typically for reuse on multiple projects within that application domain . . . [Object-oriented domain analysis is] the identification, analysis, and specification of common, reusable capabilities within a specific application domain, in terms of common objects, classes, subassemblies, and frameworks . . .

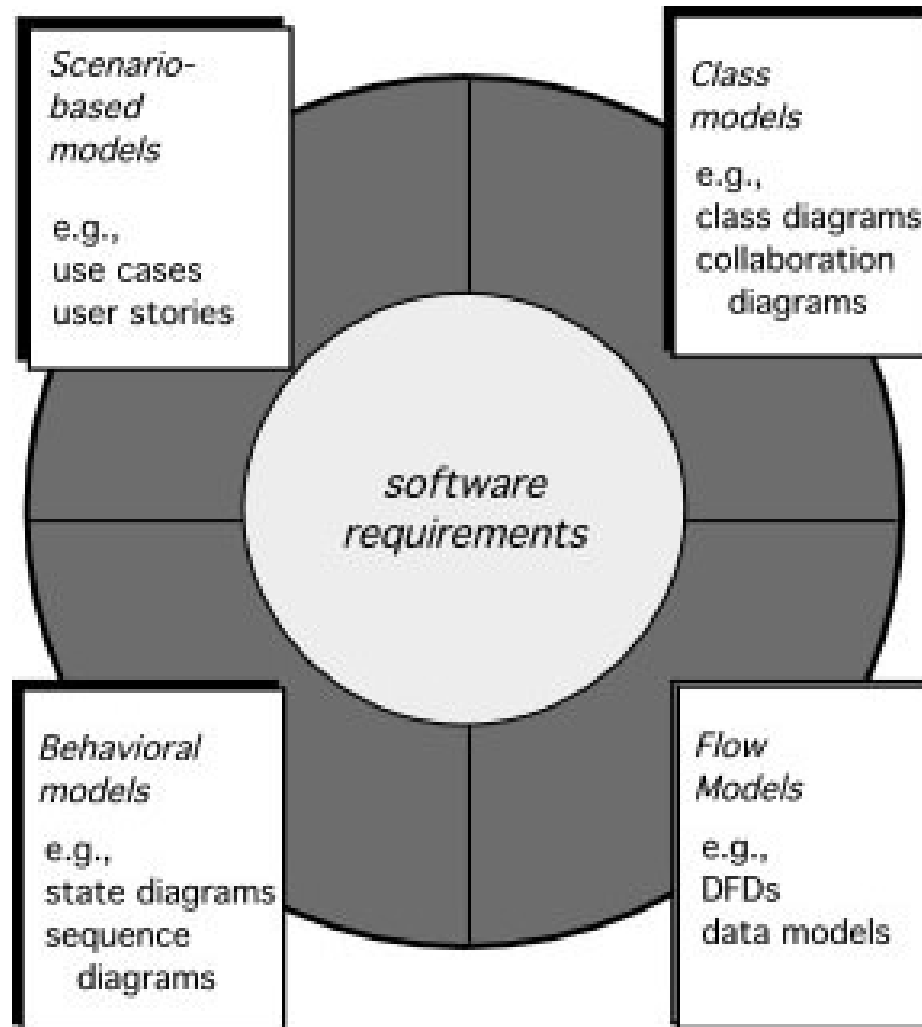
Donald Firesmith

软件领域分析是识别、分析和规范来自特定应用领域的通用需求，通常用于该应用领域内多个项目的复用。[面向对象的领域分析是]在特定应用领域内识别、分析和规范通用、可复用能力，涉及通用对象、类、子组件和框架。

Domain Analysis

- ❖ **Define the domain to be investigated.**
 - ❖ **Collect a representative sample of applications in the domain.**
 - ❖ **Analyze each application in the sample.**
 - ❖ **Develop an analysis model for the objects.**
- 定义要研究的领域。
 - 收集该领域应用的代表性样本。
 - 分析样本中的每个应用程序。
 - 为对象开发分析模型。

Elements of Requirements Analysis



Scenario-Based Modeling

“[Use-cases] are simply an aid to defining what exists outside the system (actors) and what should be performed by the system (use-cases).” Ivar Jacobson

(1) What should we write about?

(2) How much should we write about it?

(3) How detailed should we make our description?

(4) How should we organize the description?

“[用例]只是定义系统外部存在的内容（演员）以及系统应执行的内容（用例）的辅助工具。”——Ivar Jacobson

1. 我们应该写什么？
2. 我们应该写多少？
3. 我们应该使描述多详细？
4. 我们应该如何组织描述？

创建初步用例——写些什么？

Create the initial use case——

What to Write About?

- ❖ **Inception and elicitation**——provide you with the information you'll need to begin writing use cases.
- ❖ **Requirements gathering meetings, QFD（质量功能部署）, and other requirements engineering mechanisms** are used to
 - identify stakeholders
 - define the scope of the problem
 - specify overall operational goals
 - establish priorities
 - outline all known functional requirements, and
 - describe the things (objects) that will be manipulated by the system.
- ❖ To begin developing a set of use cases, **list the functions or activities performed by a specific actor.**

创建初步用例——写些什么？

Create the initial use case——

What to Write About?

- 概念和引导——提供您开始编写用例所需的信息。
- 需求收集会议、QFD（质量功能展开）和其他需求工程机制用于
 - 识别利益相关者
 - 定义问题范围
 - 指定总体操作目标
 - 确定优先级
 - 列出所有已知功能需求，并
 - 描述系统将处理的事物（对象）。
- 开始开发一组用例，列出特定演员执行的功能或活动。

Use-Cases

- ❖ a scenario that describes a “thread of usage” for a system
- ❖ *actors* represent roles people or devices play as the system functions
- ❖ *users* can play a number of different roles for a given scenario

- 描述系统“使用线程”的场景
- 演员代表系统功能中人们或设备所扮演的角色
- 用户可以在特定场景中扮演不同的角色

开发用例

Developing a Use-Case

- ❖ What are the **main tasks or functions** that are performed by the actor?
- ❖ What **system information** will the actor acquire, produce or change?
- ❖ Will the actor have to inform the system about **changes in the external environment**?
- ❖ What **information** does the actor **desire** from the system?
- ❖ Does the actor wish to be informed about **unexpected changes**?
 - 演员执行的主要任务或功能是什么?
 - 演员将获取、生成或更改哪些系统信息?
 - 演员是否需要通知系统外部环境的变化?
 - 演员希望从系统中获取什么信息?
 - 演员希望被告知哪些意外的变化?

SafeHome | 开发另一个初始用户场景

[场景] 会议室，第二次需求收集会议中。

[人物] Jamie Lazar 和 Ed Robbins，软件团队成员；Doug Miller，软件工程经理；三个市场营销人员；一个产品工程代表；一个会议主持人。

[对话]

主持人：现在是我们开始讨论 SafeHome 监视功能的时候了，让我们为访问监视功

能开发一个用户场景。

Jamie：谁在其中扮演参与者的角色？

主持人：我想 Meredith（市场营销人员）已经在该功能上进行了一些工作，你来试试这个角色吧。

Meredith：你想采用我们上次用的方法，是吗？

主持人：是的，同样的方法。

Meredith：好的，很明显，开发监视功能的理由是允许房主远距离检查房屋、记录并回放捕获的录像……就是这样。

Ed：我们采用压缩的方法存储图像吗？

主持人：好问题，但是我们现在先不考虑实现的问题，Meredith，你说呢？

Meredith：好的，这样对于监视功能基本上就有两部分……第一部分是配置系统，包括布置建筑平面图——我们需要工具来帮助房主做这件事，第二部分是实际的监视功能本身。因为布局是配置活动的一部分，所以我将重点集中在监视功能。

主持人（微笑）：抢先说出我想说的话。

Meredith：哦……我希望通过电脑或通过 Internet 访问监视功能。我的感觉是 Internet 访问可能使用的频率更高一些。

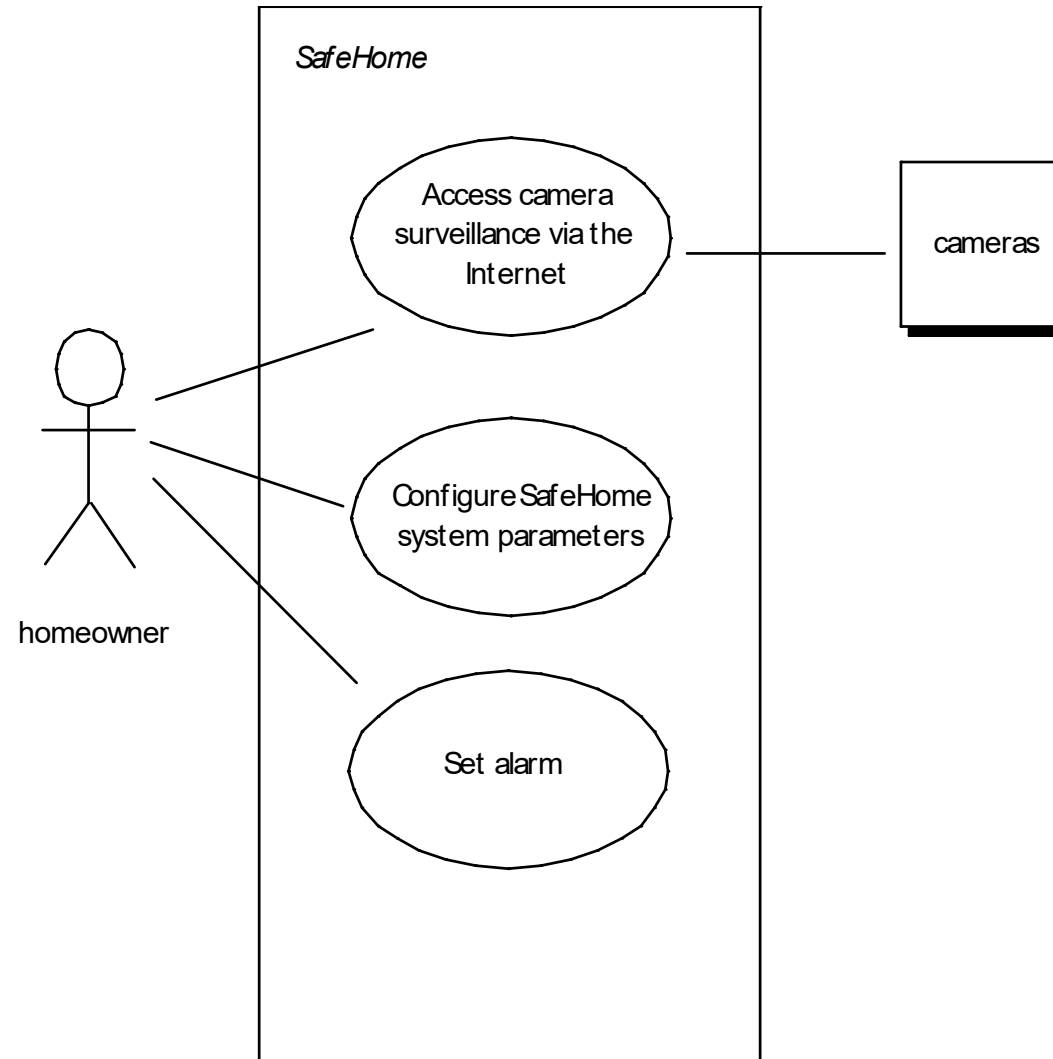
不管怎样，我希望能够在计算机上和控制面板上显示摄像机图像并移动某个摄像机镜头。在房屋平面设计图上可以选择指定摄像机，我希望可以有选择地记录摄像机输出和回放摄像机输出，我还希望能够使用特殊的密码阻止对某个或多个摄像机的访问。希望有支持小窗口显示形式的选项，即从所有的摄像机显示图像，并能够选择某一个进行放大。

Jamie：那些叫作缩略视图。

Meredith：对，然后我希望从所有摄像机获得缩略视图。我也希望监视功能的接口和所有其他的 SafeHome 接口有相同的外观和感觉。

主持人：干得好，现在，让我们更详细地讨论这个功能……

Use-Case Diagram



精炼初始用例——写多少内容？

Refine the initial use case——

How Much to Write About?

- ❖ As further conversations with the stakeholders progress, the requirements gathering team develops use cases for each of the functions noted.
- ❖ In general, **use cases are written first in an informal narrative fashion**
- ❖ If more formality is required, the same use case is rewritten using a structured format similar to the one proposed.

- 随着与利益相关者的进一步交流，需求收集团队为每个记录的功能开发用例。
- 一般而言，用例首先以非正式叙述的方式编写。
- 如果需要更多的形式化，使用类似提议的结构化格式重新编写相同的用例。

Use case:通过互联网访问摄像机监视设备

参与者：房主

- 1) 房主登录safehome网站
- 2) 房主输入账号
- 3) 房主输入两个密码
- 4) 系统显示所有的主要功能按钮
- 5) 房主在按钮中选择“监视”
- 6) 房主选择“选取摄像机”
- 7) 系统显示房屋的平面图
- 8) 房主从平面图中选择某摄像机图标
- 9) 房主选择“视图”按钮
- 10) 系统显示一个由摄像机编号确定的视图窗口
- 11) 系统在视图窗口内以每秒一帧的速度输出视频

异常情况

Exceptions

- ❖ Describe situations (failures or user choices) that cause the system to exhibit unusual behavior.
- ❖ Brainstorming should be used to derive a reasonably complete set of exceptions for each use case.
- ❖ Are there cases where a validation function occurs for the use case?
 - Are there cases where a supporting function (actor) fails to respond appropriately?
 - Can poor system performance result in unexpected or improper use actions?
- ❖ **Handling exceptions may require the creation of additional use cases.**

- 描述导致系统表现出异常行为的情况（故障或用户选择）。
- 是否有验证功能发生的用例？
- 应使用头脑风暴来为每个用例推导出的一套合理完整的异常情况。
- 是否存在支持功能（角色）未能适当响应的情况？
- 处理异常情况可能需要创建额外的用例。
- 系统性能差是否会导致意外或不当的使用行为？



Use-Case Template for Surveillance

Use-case: Access camera surveillance—display camera views (ACS-DCV).

Homeowner.

Primary actor:

Goal in context:

To view output of camera placed throughout the house from any remote location via the Internet.

Preconditions:

System must be fully configured; appropriate user ID and passwords must be obtained.

Trigger:

The homeowner decides to take a look inside the house while away.

Scenario:

1. The homeowner logs onto the *SafeHome Products* Web site.
2. The homeowner enters his or her user ID.
3. The homeowner enters two passwords (each at least eight characters in length).
4. The system displays all major function buttons.
5. The homeowner selects "surveillance" from the major function buttons.
6. The homeowner selects "pick a camera."
7. The system displays the floor plan of the house.
8. The homeowner selects a camera icon from the floor plan.

9. The homeowner selects the "view" button.
10. The system displays a viewing window that is identified by the camera ID.
11. The system displays video output within the viewing window at one frame per second.

Exceptions:

1. ID or passwords are incorrect or not recognized—see use-case: "validate ID and passwords."
2. Surveillance function not configured for this system—system displays appropriate error message; see use-case: "configure surveillance function."
3. Homeowner selects "view thumbnail snapshots for all cameras"—see use-case: "view thumbnail snapshots for all cameras."
4. A floor plan is not available or has not been configured—display appropriate error message and see use-case: "configure floor plan."
5. An alarm condition is encountered—see use-case: "alarm condition encountered."

Priority:

Moderate priority, to be implemented after basic functions.

When available:

Third increment.

Frequency of use:

Infrequent.

Channel to actor: Via PC-based browser and Internet connection to *SafeHome* Web site.

Secondary actors: System administrator, cameras.

Channels to secondary actors:

1. System administrator: PC-based system
2. Cameras: wireless connectivity

Open issues:

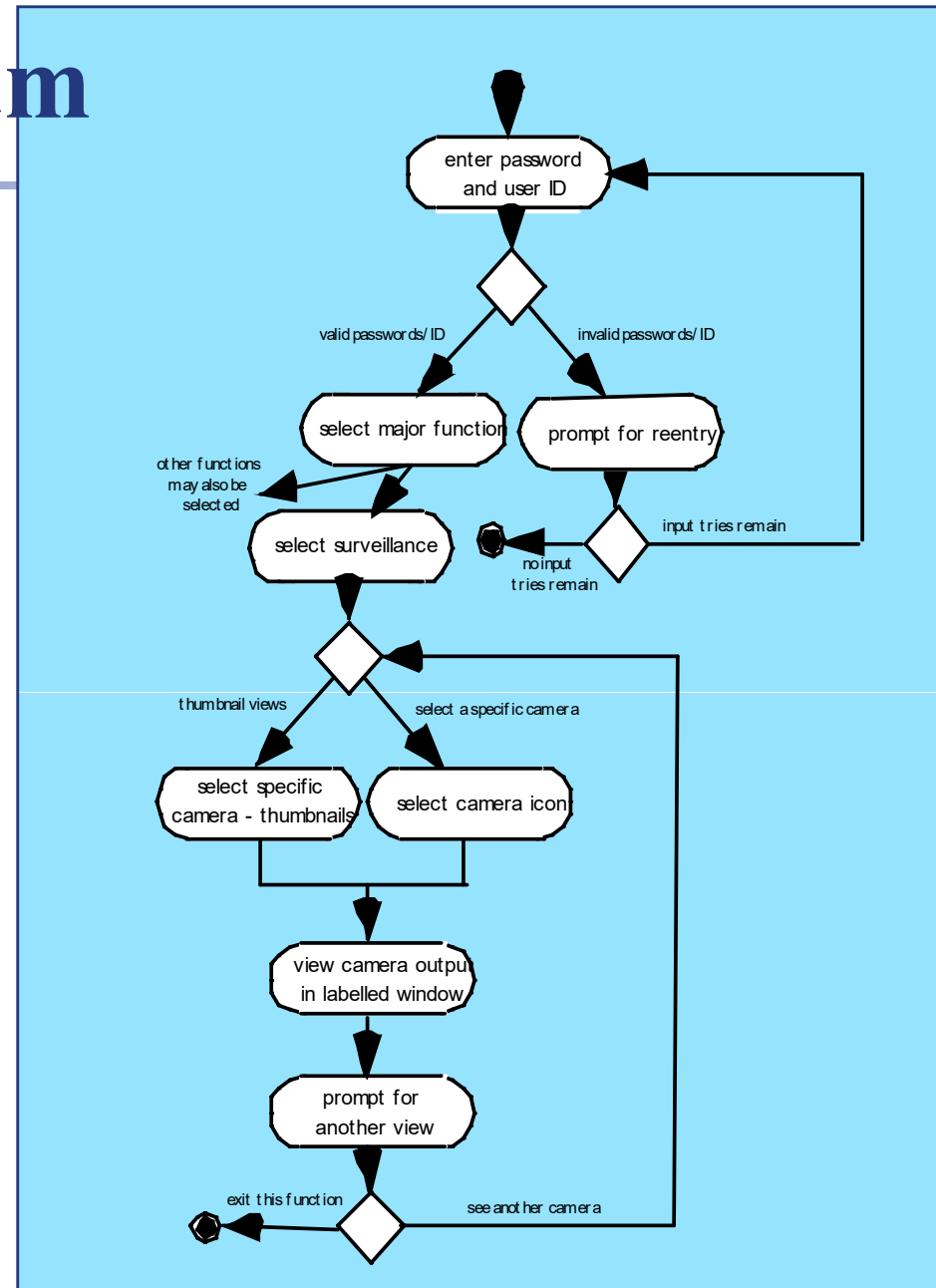
1. What mechanisms protect unauthorized use of this capability by employees of the company?

2. Is security sufficient? Hacking into this feature would represent a major invasion of privacy.
3. Will system response via the Internet be acceptable given the bandwidth required for camera views?
4. Will we develop a capability to provide video at a higher frames-per-second rate when high bandwidth connections are available?

Activity Diagram

Supplements the use case by providing a graphical representation of the flow of interaction within a specific scenario

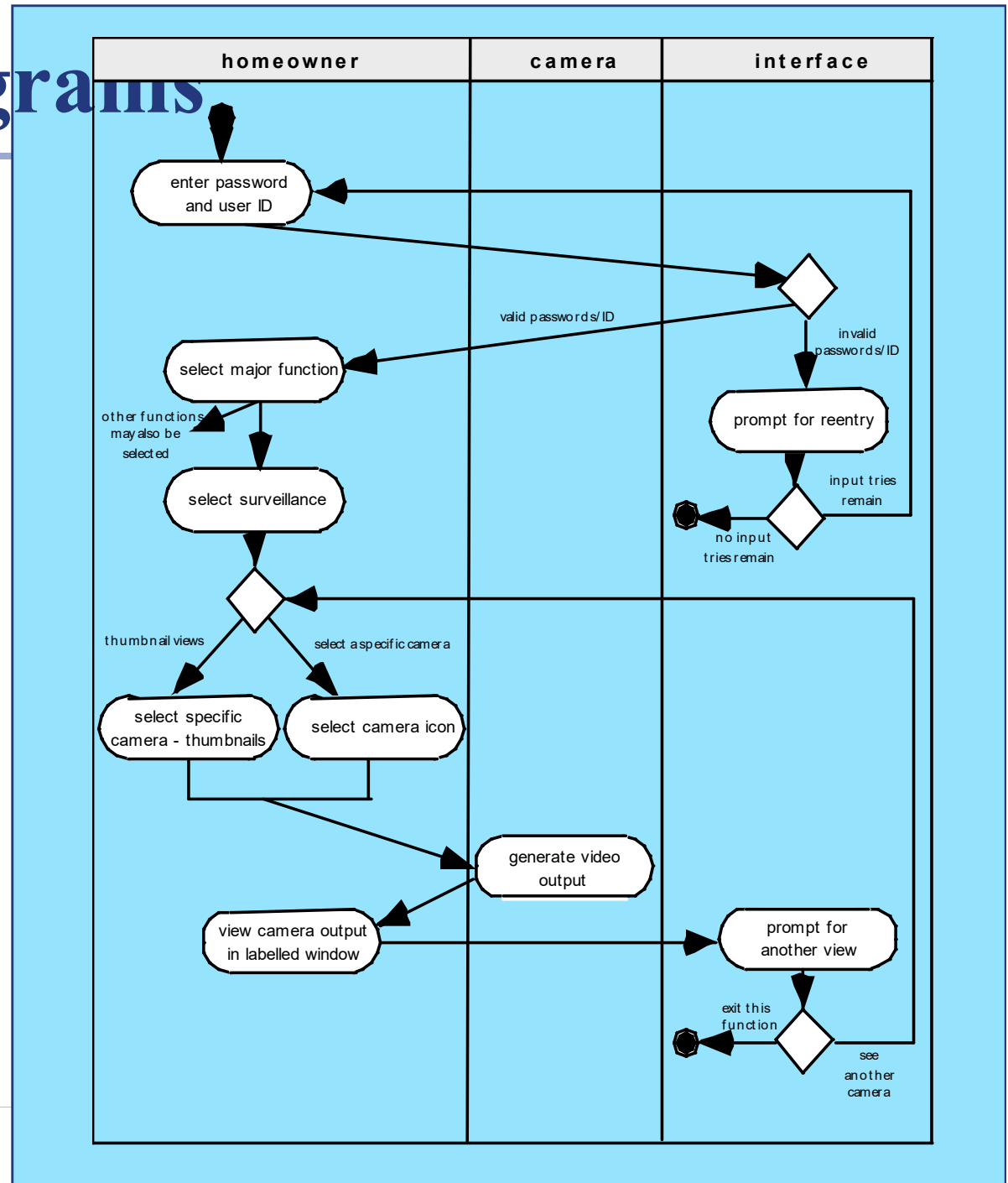
通过提供特定场景中交互流程的图形表示来补充用例。



Swimlane Diagrams

Allows the modeler to represent the flow of activities described by the use-case and at the same time indicate which actor (if there are multiple actors involved in a specific use-case) or analysis class has responsibility for the action described by an activity rectangle

允许建模者表示用例描述的活动流程，并同时指示哪个角色（如果有多个角色参与特定用例）或分析类对活动矩形描述的操作负责。



Reviewing a Use-Case

- ❖ **Use-cases are written first in narrative (叙述) form and mapped to a template if formality is needed.**
- ❖ **Each primary scenario should be reviewed and refined to see if alternative interactions are possible.**
 - **Can the actor take some other action at this point?**
 - **Is it possible that the actor will encounter an error condition at some point? If so, what?**
 - **Is it possible that the actor will encounter some other behavior at some point? If so, what?**
- 用例首先以叙述（叙述）形式编写，并在需要时映射到模板中。
- 每个主要场景都应进行审查和完善，以查看是否有可能进行替代交互。
 - 角色在此时可以采取其他行动吗？
 - 是否可能角色在某个时间点会遇到错误情况？如果是，是什么？
 - 是否可能角色在某个时间点会遇到其他行为？如果是，是什么？



Southeast University



Thank You !

lliao@seu.edu.cn